

Building an Affective Database for Emotion Detection from Natural Bangla Text

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Abstract. Emotion detection is a task within the field of Affective Computing, which focuses on designing systems and technologies capable of recognizing, interpreting, processing, and simulating human emotions. Building an affective database for Bangla text can have numerous benefits, including improving the accuracy of emotion detection, enhancing natural language processing, and providing a deeper comprehension of how people express emotions in Bangla culture. Identifying emotions from textual data and creating a consistent, reliable, and unbiased emotion dataset pose significant difficulties, especially in low-resourced languages like Bangla. This study addresses the problem of emotion detection in Bangla texts. An emotion dataset was meticulously annotated based on Paul Ekman’s six basic emotions, employing techniques to ensure dataset consistency, reliability, and impartiality. The study achieved an average Cohen’s Kappa score of 87%, indicating nearly perfect agreement between the annotator and the participants. Additionally, various machine learning models and word embeddings were evaluated for emotion detection. The findings demonstrate that Logistic Regression, SVM, and Naïve Bayes yielded the best performance, with Logistic Regression (BOW + Unigram) achieving an accuracy of 68.8% and a macro F1-score of 65%.

Keywords: Bangla NLP · Textual Emotion Classification · Text Corpus · Annotated Dataset · Machine Learning Techniques.

1 Introduction

Emotion Detection is precisely identifying human emotion based on contextual data [4]. Emotion detection is a subfield of sentiment analysis that focuses on the extraction and analysis of emotions. Emotion detection from text is challenging [8, 21] because it requires understanding the context, subtleties, and nuances of language. Text can be ambiguous; various people convey their emotions differently based on their culture, personalities, and personal experiences. Moreover, emotions are often expressed indirectly or implicitly in text, making it difficult for computers to detect them accurately.

In the case of the Bangla language, the complexity is amplified due to the unique linguistic and cultural characteristics of the language [20], as well as the

lack of labelled data for training and evaluating models. Despite these challenges, advancements in NLP and deep learning have made Bangla emotion recognition possible, and there is growing interest in the topic due to its potential applications in customer service, sentiment analysis, mental health evaluations, and conversational AI systems.

Building an emotion dataset is complex because emotions are subjective and can vary between individuals and contexts [23]. It requires annotators who can identify and agree on the emotion conveyed in a text. Also, creating a diverse and representative dataset that includes a range of emotions and contexts is challenging, especially for languages like Bangla. This article explores the methods for overcoming these challenges by proposing the construction of a consistent, unbiased, and reliable dataset for emotion classification. It examines several machine learning methods to detect emotions from natural Bangla text. This research establishes a foundational contribution to emotion detection in the Bangla language, emphasizing the importance of creating consistent and reliable datasets for future investigations.

2 Related Works

Emotion models are the foundation of ED systems [1], with two categories of models: discrete emotion models (DEMs) and dimensional emotion models (DiEMs). DEMs place emotions into distinct categories, such as Ekman’s six basic emotions, Plutchik’s eight emotions, and OCC’s (Ortony/Clore/Collins) 22 categories. DiEMs place emotions in a spatial space based on arousal and valence, such as Russell’s 2D model and Plutchik’s 2D wheel of emotions.

Emotion detection from text in the Bangla language is a relatively new area of research in NLP. However, emotion recognition from Bangla text has received more focus recently in the field of Natural Language Processing (NLP). The discrete emotion model underpins most Bangla emotion datasets.

Tripto et al. [24] gathered data from YouTube user comments on Bangla, English, and Romanized Bangla. Two datasets are for sentiment analysis and one for emotion detection. To detect sentiment and emotions, the authors developed a deep learning-based model that combines Long Short Term Memory and CNN with an embedding layer. This model’s accuracy on an emotion database with six basic emotions was 59.2%. (anger, disgust, fear, joy, sadness, and surprise).

Rahman et al. [17] collected and annotated a corpus of Bangla text that included user comments from several Facebook groups about socioeconomic and political topics and extracted basic emotions based on the Paul Ekman emotion model. SVM provided the best results in their experiment, reaching an accuracy score of 52.98%

Azmin et al. [3] proposed a Multimodal Naive Bayes (NB) classifier that incorporates several features such as stemmers, POS taggers, n-grams, and term frequency-inverse document frequency (TF-IDF). They conducted their experiments using data from Facebook comments [17] and achieved an accuracy of 78.6%.

Pal et al. [14] created Anubhuti, the largest text corpus consisting of 32,124 sentences, for analyzing the emotions portrayed by Bangla short story writers. They addressed some challenges that a low-resource language like Bangla poses in the raw data collecting and annotation process.

For the classification of Bangla text in various areas, including sentiment analysis, emotion detection, news categorization, and author identification, Alam et al. [2] proposed fine-tuned multilingual transformer models. Their proposed XLM-RoBERTa model achieved 72.7% accuracy on the YouTube emotion detection dataset [24].

Parvin et al. [15] proposed an ML-based ensemble method for categorizing Paul Ekman’s six main textual emotions. They trained and tested eight machine learning algorithms using Bag of Words (BoW) and TF-IDF feature extraction on an emotion corpus of 8047 Bangla texts. Their ensemble technique, LR, RF, and SVM, had the greatest weighted f1-score of 62.39%.

Hoq et al. [7] created four models that combine a CNN layer and an LSTM layer with different Word Embeddings, such as Embedding Layer, Word2Vec, Global Vectors, and Continuous Bag of Words. Their dataset was created using Facebook comments consisting of three basic emotions; happiness, anger, and sadness. The best results were obtained using a CNN-LSTM hybrid Word2Vec embedding layer, with an accuracy of 90.49% and an F1 score of 92.83% .

Combining the BiLSTM recurrent neural network and SVM-NB classification with pre-training with the ALBERT model, Zihao Ye et al. [26] proposed a novel method for textual emotion recognition and increased emotion recognition accuracy. ChineseNlpCorpus, NLP&CC 2014, and WEC, three frequently used Chinese datasets, were used. Their proposed model outperformed competing models like LSTM, BiLSTM, and BERT-BiLSTM by reaching accuracy rates of 92.96%, 91.27%, and 93.11%, on those datasets, respectively.

As emotion in texts can be ambiguous [12], it is essential to build a proper and consistent dataset to ensure the accuracy of emotion detection models. However, the existing literature does not discuss the methods for building an emotion dataset in-depth. Therefore, in this work, we propose a comprehensive methodology for constructing an emotion dataset in the Bangla language. By presenting our approach, we aim to contribute to the advancement of research in emotion detection from Bangla text.

3 Challenges in Bangla Emotion Detection

Both emotion dataset creation and emotion detection are challenging tasks in NLP for several reasons:

- **Nature of Bangla sentences:** Small texts can convey emotions even without extensive context or background information through the use of emotive language, such as descriptive words and phrases, as well as syntax and punctuation. The tone, style, and structure of a text can also provide cues about the emotions it conveys. For example, a text written in

a sad or melancholic tone is likely to convey sadness, while a text written in an angry or frustrated tone is likely to convey anger. Additionally, the choice of words and phrases can also help to convey emotions and can explicitly describe the sentiment of a text, such as the use of words like "আনন্দ", "কষ্ট", "রাগ", "ছিঃ ছিঃ", "ধ্যাত", "মূর্খ".

- **Cultural and Linguistic Variation:** Emotions and sentiments are culturally and linguistically specific, meaning that the same words or phrases can have different meanings and connotations in different languages and cultures. This makes it difficult to create datasets that are relevant and generalizable across different cultures.
- **Last few lines:** The last few lines of a story can convey the emotion of the whole story by emphasizing or summarizing the main emotional events or themes that took place throughout the story. These lines often provide a resolution or conclusion to the story’s conflicts, which can significantly influence the reader’s overall emotional interpretation of the story. Additionally, the tone, wording, and language used in the final lines can also contribute to the conveyed emotion, helping to solidify the reader’s understanding of the story’s emotional impact. This can be challenging for the emotion detection task, as the model may not have access to enough information to identify the emotion accurately.
- **Annotation Bias:** The process of annotating texts with emotions is subjective and can be influenced by annotators’ own cultural and linguistic backgrounds, experiences, and biases. This can result in annotated datasets that are inconsistent and unreliable.
- **Limited Labeled Data:** Creating high-quality labelled datasets for emotion detection requires a large amount of annotated data, which can be challenging to obtain and time-consuming to annotate.

4 Methodology

This study collected and annotated a considerable corpus of emotion-labelled text data to employ machine learning algorithms for emotion detection. First, a vast corpus of text data was annotated with emotions including happiness, anger, sadness, and others. Next, machine learning methods were used to detect emotions. These algorithms were trained on annotated data and tested on a held-out set. Accuracy, precision, recall, and F1-score were evaluated to find the best emotion-detecting algorithm.

4.1 Building an Affective Database

The data collection for the emotion detection task from Bangla texts involved gathering a large corpus of publicly available text data from various sources on the internet. The collected data was pre-processed to remove any irrelevant information and to ensure consistency in terms of language, format, and encoding. This pre-processed data was then annotated with emotions such as joy, anger,

sadness, and others. The annotation process involved assigning a label to each piece of text indicating the predominant emotion conveyed by the text. The preparation of dataset is done in 3 stages [23], as shown in Fig. 1:

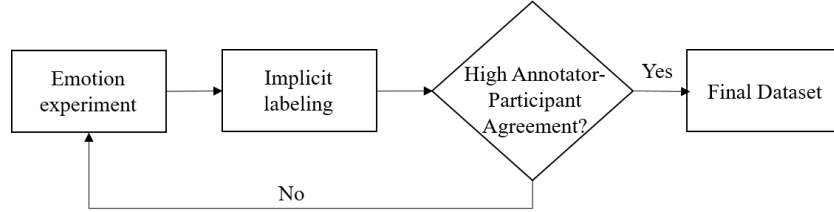


Fig. 1. Dataset preparation workflow.

Emotion Experiment This stage involves the annotator finding emotion texts or sentences from a set of corpora and labelling each of them by the predominant emotion. Texts are labelled according to Ekman’s six basic emotion labels (Happiness, Sadness, Anger, Disgust, Surprise, and Fear). This stage involves 3 basic steps:

1. **Data collection:** Gather a large corpus of Bangla text data from publicly available sources such as blogs, Facebook and YouTube posts and comments, poems, stories, songs, plays, and novels.
2. **Pre-processing:** Clean and pre-process the collected data to remove irrelevant information, correct any errors, and ensure consistency in terms of language, format, and encoding.
3. **Annotation:** Assign a label to each piece of pre-processed text indicating the predominant emotion the text conveys. The annotators follow basic annotation guidelines to ensure consistency and accuracy.

Table 1. Examples of some keywords associated to particular emotion classes.

Emotion	Keywords	Example
Happiness	শুভ, খুশি, আনন্দ, ভালো, শান্তি	আহা কি আনন্দ, পরীক্ষা শেষ হয়েছে
Sadness	কষ্ট, খারাপ, হতাশা, বিষম	আকাশ বিষম, ডেসু আর বন্যা শহরে আবার ফিরে এসেছে
Anger	রাগ, ক্রোধ, ক্ষোভ	আমাদের ক্ষোভ সিস্টেমের বিরুদ্ধে
Disgust	ঘৃণা, অরুচি, ফাউল, জঘন্য	ওহ পানির স্বাদ জঘন্য
Surprise	হতবাক, অবাক, বিস্মিত, বাহ্	আমি হতবাক হয়ে গিয়েছি এই জল তৈরির কারবার দেখে
Fear	উদ্বেগ, ভয়, আতঙ্ক, শঙ্কিত	ভবিষ্যৎ নিয়ে অত্যধিক সংশয় আর ভয় হচ্ছে

To ensure consistency and accuracy in the annotation step, the following guidelines or rules should be followed:

- **Emotion definitions:** Annotator must understand the meaning and nuances of each emotion. Some keywords may be associated to particular emotion classes [Table 1]. A sentence or sub-sentence can be added if [Table 2],
 - it is semantically acceptable (contains a context or meaning)
 - contains a predominant emotion
- **Multiple annotators:** Use multiple annotators to annotate each piece of text and take the majority vote as the final label. This helps to ensure that the annotations are accurate and eliminates the possibility of a single annotator’s biases affecting the dataset.

Table 2. Examples of sentences or sub-sentences that can or can’t be added to the dataset.

Sentence	Label/ Remarks	Added
হায় হায় কি বলেন এসব	বিস্ময়	Yes
ক্ষতি নেই	Not a complete sub-sentence	No
এপাশে ওর কেউ নেই	দুঃখ	Yes
হায় হায়!	Semantically acceptable but ambiguous	No
চোপ!	রাগ	Yes
যে পিতা সন্তানের লাশ সনাক্ত করতে ভয় পায় আমি তাকে ঘৃণা করি	বিতৃষ্ণা	Yes

Implicit Labelling In this stage, samples of emotion-related sentences are shown to the participants to assess the reliability of the annotator. This stage has 3 steps [Fig. 2].

1. Emotion sentences are shown to the participants without any emotion label.
2. Same emotion sentences are shown to the participants, this time with emotion labels.
3. Participants are asked if they agree with the labelling.

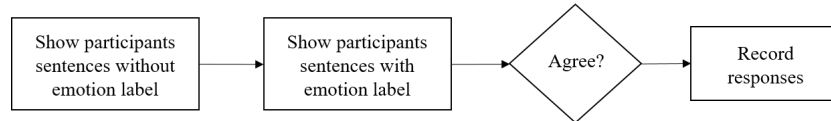


Fig. 2. Workflow of the implicit labelling stage.

Annotation Reliability One of the challenges of emotion detection dataset creation is annotation bias. The process of annotating texts with emotions is subjective and can be influenced by annotators’ own cultural and linguistic backgrounds, experiences, and biases. This can result in annotated datasets that are inconsistent and unreliable.

In the third stage [Fig. 1], annotator reliability is calculated using Cohen’s Kappa [Eqn. 1]. Cohen’s Kappa [5, 25] is a statistical measure used to quantify the agreement between two annotators or raters who are coding a set of items, such as texts, for a specific attribute, such as emotion. If the annotation of a sentence isn’t reliable enough (e.g. kappa score < 0.8), then that sentence is annotated again or discarded.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (1)$$

where p_o is the observed agreement between the annotators, and p_e is the expected agreement between the annotators under the assumption of random chance.

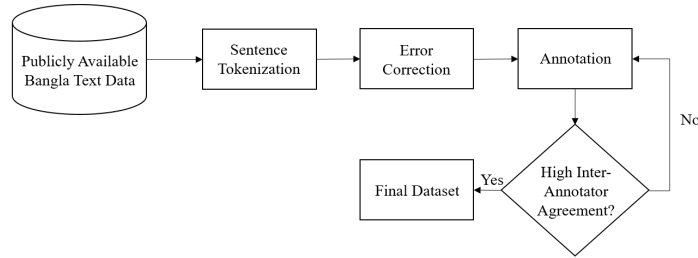


Fig. 3. Summary of the dataset creation process.

4.2 Emotion Detection

After preparing the emotion dataset, the next stage is detecting emotion from texts. The goal in this stage is to train machine learning models to identify the emotions in the text data automatically. The preprocessing stage involves text cleaning, word tokenization, stopwords removal, and stemming. Feature representation involves techniques like word embedding (i.e. Bag of Words, TF-IDF [9], Word2Vec [10, 11]) and word n-gram. The data is then split into training and testing sets, and different machine learning models such as Naive Bayes [18], Logistic Regression [6], KNN [16], SVM [13], Decision Tree [19] and Random Forest [22] is used. The final model is then evaluated and fine-tuned before being deployed in a production environment for emotion detection in new Bangla texts.

Exclamation points and question marks are not removed during the text cleaning stage because they can carry important information about the sentiment or emotion expressed in a text. For example, exclamation points can indicate strong emotions such as excitement, anger, or surprise. In contrast, question marks can indicate that a text is a question and may also convey uncertainty, confusion, or scepticism. Therefore, keeping these punctuation marks can help differentiate between emotional and neutral or weakly emotional texts, and improve the performance of the machine learning models in identifying emotions.

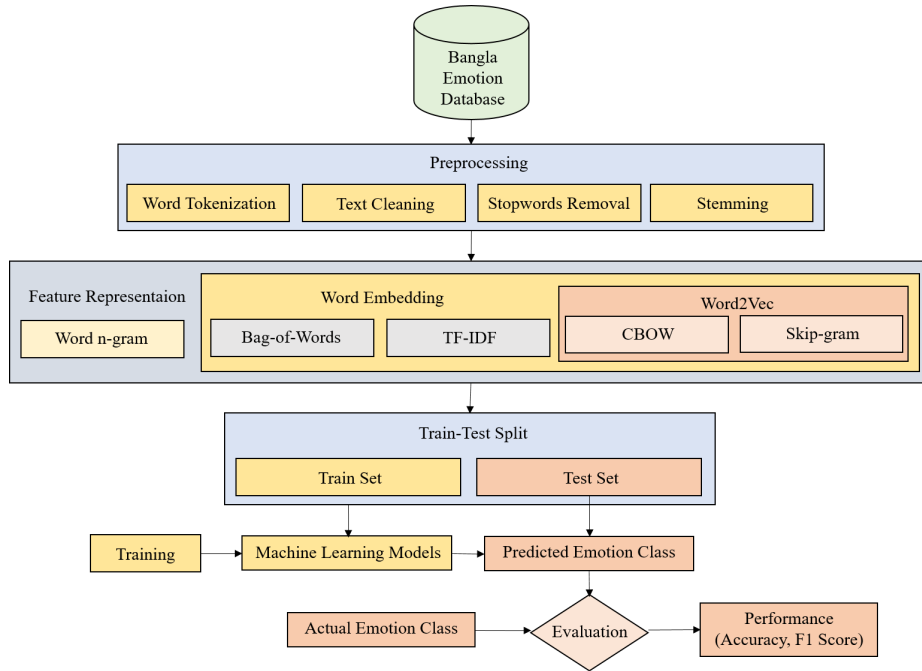


Fig. 4. Workflow of emotion detection task.

5 Experimental Analysis

5.1 Evaluation Metrics

In the dataset creation phase, annotator reliability was calculated using Cohen's Kappa [Eqn. 1]. And in the emotion detection phase, some commonly used evaluation metrics were used to measure the performance of the machine learning models on the test set.

Accuracy Accuracy measures the percentage of correctly predicted emotion labels in the test set.

$$Accuracy = \frac{\text{Number of correctly predicted emotion labels}}{\text{Total number of emotion labels}} \quad (2)$$

Precision Precision is the ratio of true positives (TP) among all the predicted positives (TP + FP). It is useful when the focus is on minimizing the false positives (FP) in the predictions.

Recall Recall measures the proportion of true positives (TP) among all the actual positives (TP + FN). It is useful when the focus is on minimizing the false negatives (FN) in the predictions.

F1-Score F1-Score is the harmonic mean of precision and recall. It is a balanced measure that takes into account both precision and recall. It is defined as follows:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

These measures determine the optimal machine learning model for emotion detection. Typically, the model with the highest F1-Score, which factors in precision and recall, is selected. Micro F1 score treats the entire dataset as a single entity and calculates precision and recall globally. It gives equal weight to each instance, regardless of its class. This means that the performance of the minority class will have the same impact as that of the majority class. A micro F1 score helps to evaluate overall performance across all classes. On the other hand, the macro F1 score determines the F1 score for every class separately before averaging them together. Regardless of how frequently a class appears in the dataset, it considers all classes identically. A macro F1 score can be helpful when the model's ability to handle minority classes needs to be evaluated and avoid the dominant class from overshadowing the evaluation.

In the Bangla text emotion detection task, some emotions may have more instances than others, resulting in an imbalanced dataset. In such situations, macro scores are favoured over micro scores because micro scores can be biased towards the majority class.

5.2 Emotion Dataset

Emotion texts in the Bangla language were collected from various publicly available sources such as quotes, blogs, Facebook and YouTube posts, poems, stories, songs, plays, and novels. 5139 unique Bangla emotion texts were collected and labelled with their predominant emotions. The distribution of collected data is shown in a bar plot where Happiness and Sadness are the most frequent emotions encountered. Some of the prose/poetries emotion texts were collected from

are "নন্দলাল"-দ্বিজেন্দ্রলাল রায়, "বহিপীর"-সৈয়দ ওয়ালীউল্লাহ, "মাসি-পিসি"-মানিক বন্দ্যোপাধ্যায়, "দেনাপাওনা"-রবীন্দ্রনাথ ঠাকুর, "লালসালু"-সৈয়দ ওয়ালীউল্লাহ and so on.

A single annotator collected and labelled all the emotion data, so to evaluate the reliability of the annotator, five participants were asked to label sets of randomly sampled texts (each containing 400 texts) from the dataset. Cohen’s Kappa scores were calculated for each class [Table 3], and the average score for each class was above 80%, indicating almost perfect agreement between the annotator and the participants. However, the scores for the Anger and Disgust classes were relatively lower, indicating that these emotions are harder to distinguish.

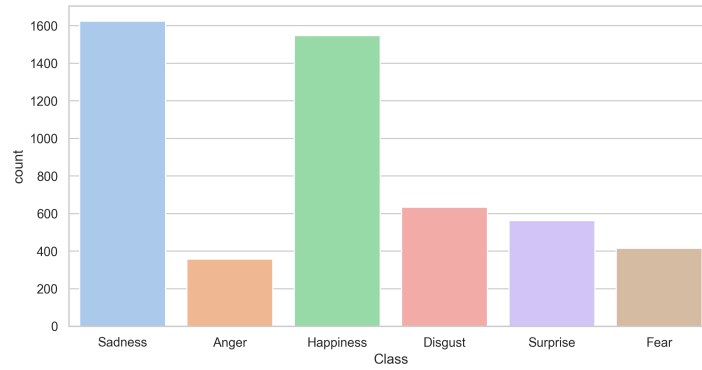


Fig. 5. Distribution of collected data in each emotion class.

Table 3. Cohen’s Kappa scores for measuring the level of agreement between the annotator and participants.

[illegible]

The average sentiment polarity was calculated for each emotion class [Fig. 6], and it was found that *Fear*, *Disgust*, and *Anger* classes have negative sentiment polarity because they are typically associated with negative reactions or feelings towards something unpleasant.

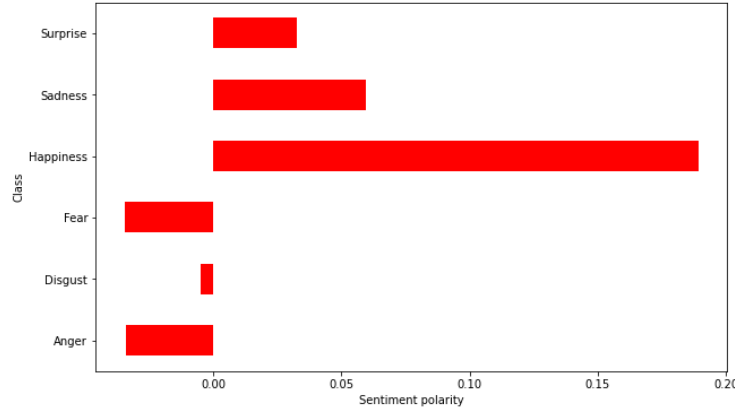


Fig. 6. Sentiment polarity of collected data for each emotion class.

Finally, figure 7 shows frequently occurring words in each emotion classes. Exclamation points (!) are primarily present in classes *Anger*, *Surprise*; and question marks (?) occurred mostly in *Surprise* and, *Disgust*.

5.3 Results of Emotion Recognition

Different machine learning models were used with various word embedding techniques for training to classify emotions. The models [Table 4] used were Naive Bayes, Logistic Regression, KNN, SVM, Decision Trees, and Random Forest. Random sampling train-test split was used. 15% of the available texts were used for the evaluation. The accuracy and macro precision of most SVM models using Bag of Words and TF-IDF feature representation were above 62% and 58%, respectively. Logistic Regression, SVM, and Naive Bayes performed better at classifying all classes than other models, with Logistic Regression (BOW + Unigram) having the highest accuracy of 68.8% and macro F1-score of 65%; SVM (TF-IDF + Bigram) having an accuracy of 67.96% and macro F1-score of 63%; and Naive Bayes (BOW + Unigram) having an accuracy of 66.78% and macro F1-score of 62%. Even though Word2Vec was expected to perform better, it did not improve the performance when coupled with the dataset because of the low amount of data.

Logistic Regression, Support Vector Machines (SVM), and Naïve Bayes outperformed other models in emotion detection due to their distinct characteristics. SVM is capable of capturing complex decision boundaries and handling

Table 4. Performance of machine learning models on collected dataset – Accuracy, Precision, Recall, and F1-score.

Model		Accuracy	Precision Macro Weighted		Recall Macro Weighted		F1-score Macro Weighted	
Naive Bayes	BOW (Unigram)	0.6678	0.76	0.71	0.57	0.67	0.62	0.66
	BOW (Trigram)	0.6594	0.73	0.69	0.56	0.66	0.60	0.65
	TF-IDF (Unigram)	0.5683	0.80	0.69	0.39	0.57	0.41	0.51
Logistic Regression	BOW (Unigram)	0.688	0.74	0.71	0.62	0.69	0.65	0.68
	BOW (Trigram)	0.6695	0.68	0.67	0.60	0.67	0.62	0.66
KNN	BOW (Unigram)	0.339	0.44	0.5	0.34	0.34	0.29	0.34
	TF-IDF (Bigram)	0.543	0.53	0.53	0.50	0.54	0.51	0.54
	Word2Vec	0.321	0.2	0.26	0.2	0.32	0.18	0.27
SVM	BOW (Bigram)	0.6273	0.78	0.70	0.54	0.63	0.58	0.61
	TF-IDF (Bigram)	0.6796	0.8	0.73	0.59	0.68	0.63	0.67
	TF-IDF (Trigram)	0.6779	0.81	0.73	0.59	0.68	0.63	0.67
	Word2Vec	0.4301	0.42	0.42	0.3	0.43	0.29	0.38
Decision Tree	BOW (Trigram)	0.597	0.62	0.61	0.55	0.60	0.57	0.59
	TF-IDF (Unigram)	0.5784	0.58	0.59	0.55	0.58	0.55	0.58
Random Forest	BOW (Unigram)	0.6425	0.76	0.71	0.58	0.64	0.61	0.63
	TF-IDF (Unigram)	0.6408	0.74	0.69	0.59	0.64	0.62	0.63
	Word2Vec	0.5	0.57	0.52	0.34	0.50	0.34	0.44

References

1. Acheampong, F.A., Wenyu, C., Nunoo-Mensah, H.: Text-based emotion detection: Advances, challenges, and opportunities. *Engineering Reports* **2**(7), e12189 (2020)
2. Alam, T., Khan, A., Alam, F.: Bangla text classification using transformers. *arXiv preprint arXiv:2011.04446* (2020)
3. Azmin, S., Dhar, K.: Emotion detection from bangla text corpus using naive bayes classifier. In: 2019 4th International Conference on Electrical Information and Communication Technology (EICT). pp. 1–5. IEEE (2019)
4. Binali, H., Wu, C., Potdar, V.: Computational approaches for emotion detection in text. In: 4th IEEE international conference on digital ecosystems and technologies. pp. 172–177. IEEE (2010)

5. Cohen, J.: A coefficient of agreement for nominal scales. *Educational and psychological measurement* **20**(1), 37–46 (1960)
6. Dreiseitl, S., Ohno-Machado, L.: Logistic regression and artificial neural network classification models: a methodology review. *Journal of biomedical informatics* **35**(5-6), 352–359 (2002)
7. Hoq, M., Haque, P., Uddin, M.N.: Sentiment analysis of bangla language using deep learning approaches. In: *International Conference on Computing Science, Communication and Security*. pp. 140–151. Springer (2021)
8. Islam, S., Roy, A.C., Arefin, M.S., Afroz, S.: Multi-label emotion classification of tweets using machine learning. In: *Proceedings of the International Conference on Big Data, IoT, and Machine Learning: BIM 2021*. pp. 705–722. Springer (2022)
9. Liu, C.z., Sheng, Y.x., Wei, Z.q., Yang, Y.Q.: Research of text classification based on improved tf-idf algorithm. In: *2018 IEEE International Conference of Intelligent Robotic and Control Engineering (IRCE)*. pp. 218–222. IEEE (2018)
10. Mikolov, T., Chen, K., Corrado, G., Dean, J.: Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781* (2013)
11. Mikolov, T., Sutskever, I., Chen, K., Corrado, G.S., Dean, J.: Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems* **26** (2013)
12. Nawrin, S., Hasan, K.A.: Emotion detection from natural text using embedding techniques. In: *Applied Informatics for Industry 4.0*, pp. 122–135. Chapman and Hall/CRC (2023)
13. Noble, W.S.: What is a support vector machine? *Nature biotechnology* **24**(12), 1565–1567 (2006)
14. Pal, A., Karn, B.: Anubhuti—an annotated dataset for emotional analysis of bengali short stories. *arXiv preprint arXiv:2010.03065* (2020)
15. Parvin, T., Hoque, M.M.: An ensemble technique to classify multi-class textual emotion. *Procedia Computer Science* **193**, 72–81 (2021)
16. Peterson, L.E.: K-nearest neighbor. *Scholarpedia* **4**(2), 1883 (2009)
17. Rahman, M., Seddiqui, M., et al.: Comparison of classical machine learning approaches on bangla textual emotion analysis. *arXiv preprint arXiv:1907.07826* (2019)
18. Rish, I., et al.: An empirical study of the naive bayes classifier. In: *IJCAI 2001 workshop on empirical methods in artificial intelligence*. vol. 3, pp. 41–46 (2001)
19. Safavian, S.R., Landgrebe, D.: A survey of decision tree classifier methodology. *IEEE transactions on systems, man, and cybernetics* **21**(3), 660–674 (1991)
20. Sen, O., Fuad, M., Islam, M.N., Rabbi, J., Masud, M., Hasan, M.K., Awal, M.A., Fime, A.A., Fuad, M.T.H., Sikder, D., et al.: Bangla natural language processing: A comprehensive analysis of classical, machine learning, and deep learning based methods. *IEEE Access* (2022)
21. Seyeditabari, A., Tabari, N., Zadrozny, W.: Emotion detection in text: a review. *arXiv preprint arXiv:1806.00674* (2018)
22. Shah, K., Patel, H., Sanghvi, D., Shah, M.: A comparative analysis of logistic regression, random forest and knn models for the text classification. *Augmented Human Research* **5**, 1–16 (2020)
23. Soleymani, M., Lichtenauer, J., Pun, T., Pantic, M.: A multimodal database for affect recognition and implicit tagging. *IEEE transactions on affective computing* **3**(1), 42–55 (2011)
24. Tripto, N.I., Ali, M.E.: Detecting multilabel sentiment and emotions from bangla youtube comments. In: *2018 International Conference on Bangla Speech and Language Processing (ICBSLP)*. pp. 1–6. IEEE (2018)

25. Viera, A.J., Garrett, J.M., et al.: Understanding interobserver agreement: the kappa statistic. *Fam med* **37**(5), 360–363 (2005)
26. Ye, Z., Zuo, T., Chen, W., Li, Y., Lu, Z.: Textual emotion recognition method based on albert-bilstm model and svm-nb classification. *Soft Computing* **27**(8), 5063–5075 (2023)